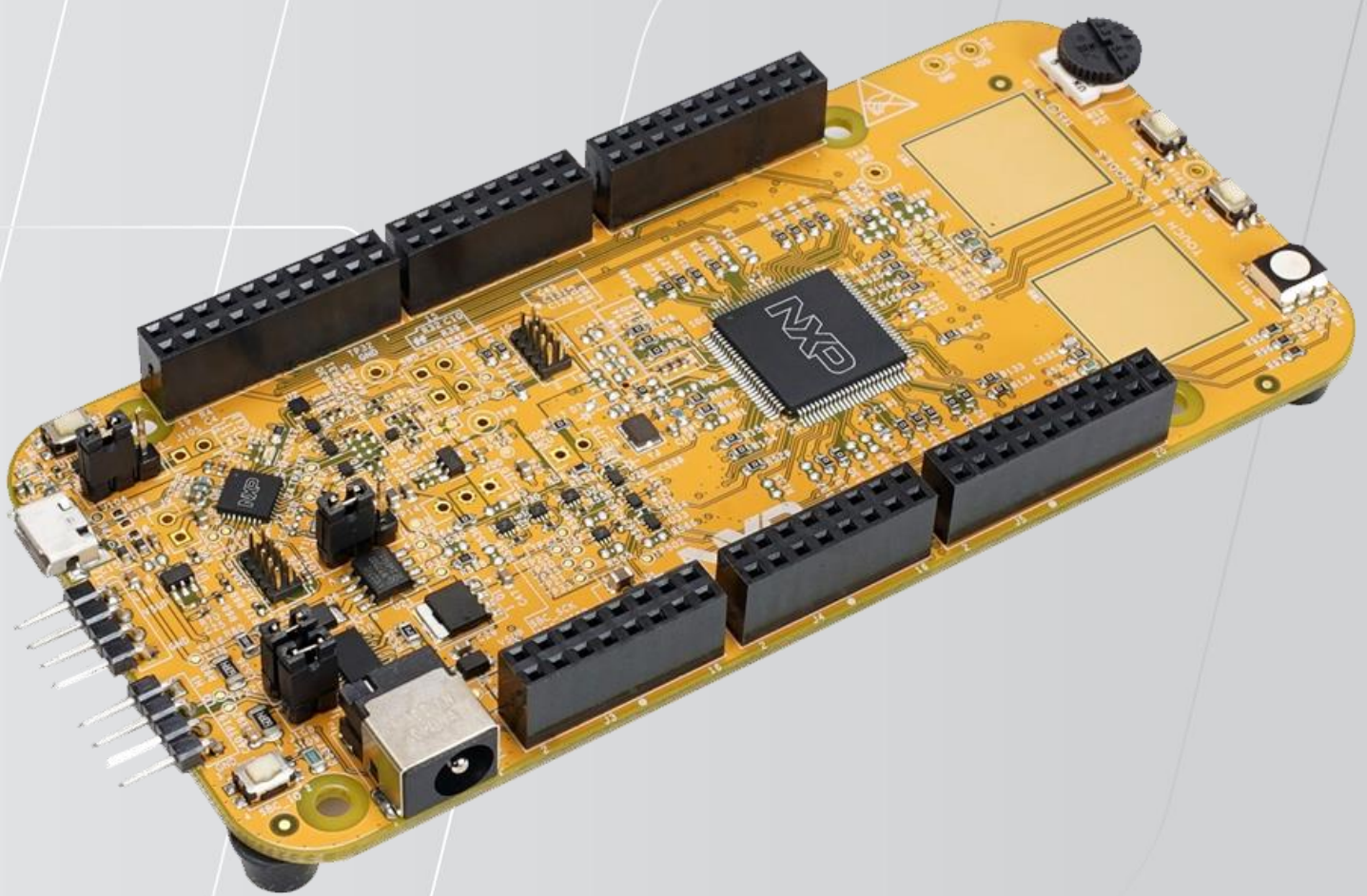
SysTick Timer



13 Sep 2023

OUTLINE

1. Overview
2. SysTick Registers
3. Operations of the SysTick timer
4. Configuration for SysTick timer

1. Overview

- The Cortex-M processors have a small integrated timer called the SysTick (System Tick) timer.
- It is integrated as a part of the NVIC and can generate the SysTick exception (exception type #15).
- The SysTick timer is a simple decrement 24-bit timer, and can run on processor clock frequency or from a reference clock frequency (normally an on-chip clock source).
- The reason for having the timer inside the processor is to help software portability. Since all the Cortex-M processors have the same SysTick timer, an OS written for one Cortex-M3/M4 microcontroller can be reused on other Cortex-M3/M4 microcontrollers.
- There are a number of considerations when using the SysTick timer:
  - The registers in the SysTick timer can only be accessed when in privileged state.
  - The reference clock might not be available in some microcontroller designs.
  - When you are using an embedded OS in your application, the SysTick timer will be used by the OS and therefore should not be used by the application tasks.
  - The SysTick Timer stops counting when the processor is halted during debugging.
  - Depending on the design of the microcontroller, the SysTick timer may stop in certain sleep modes.

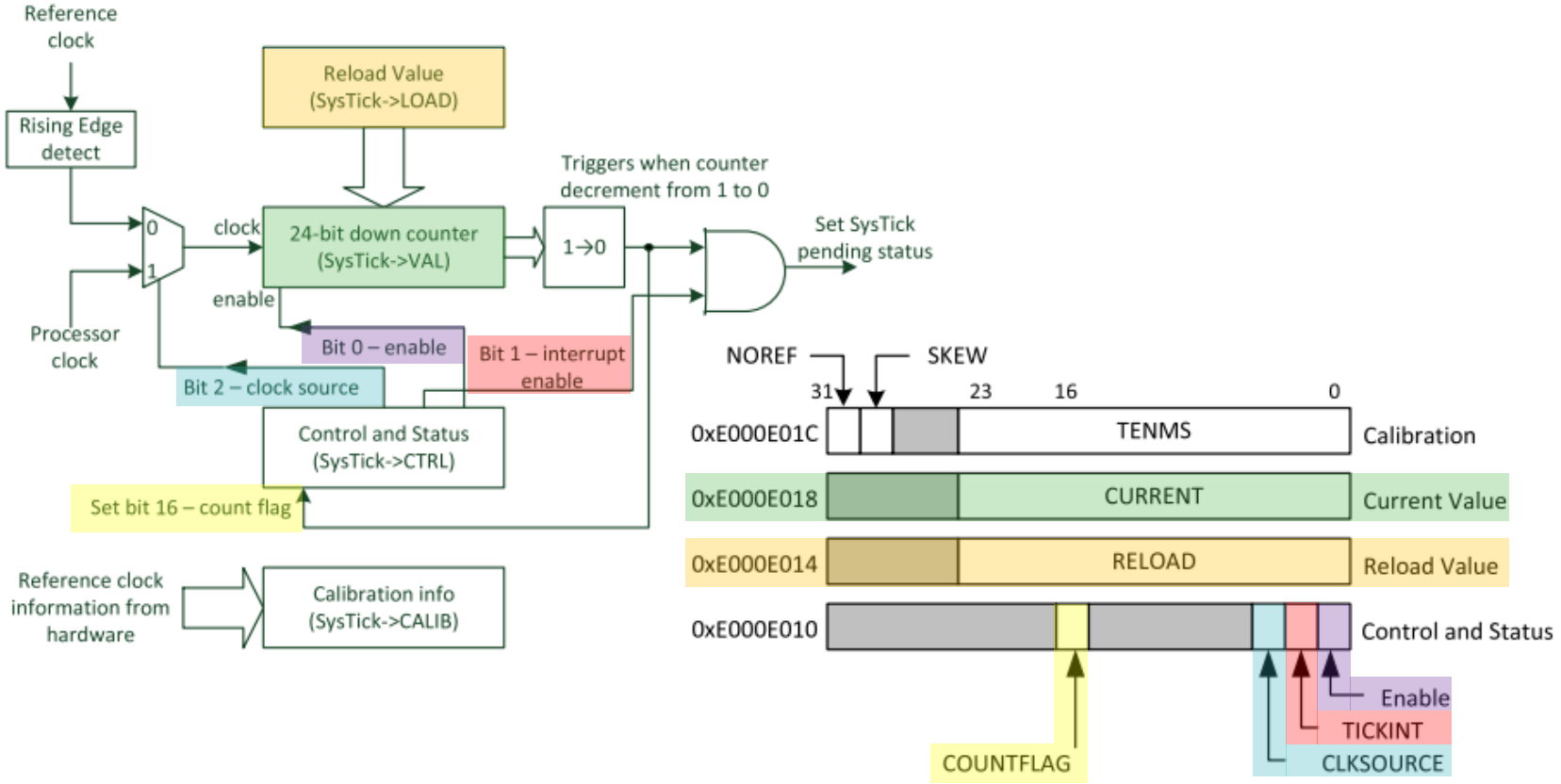
2. SysTick Registers

- The SysTick timer contains four registers

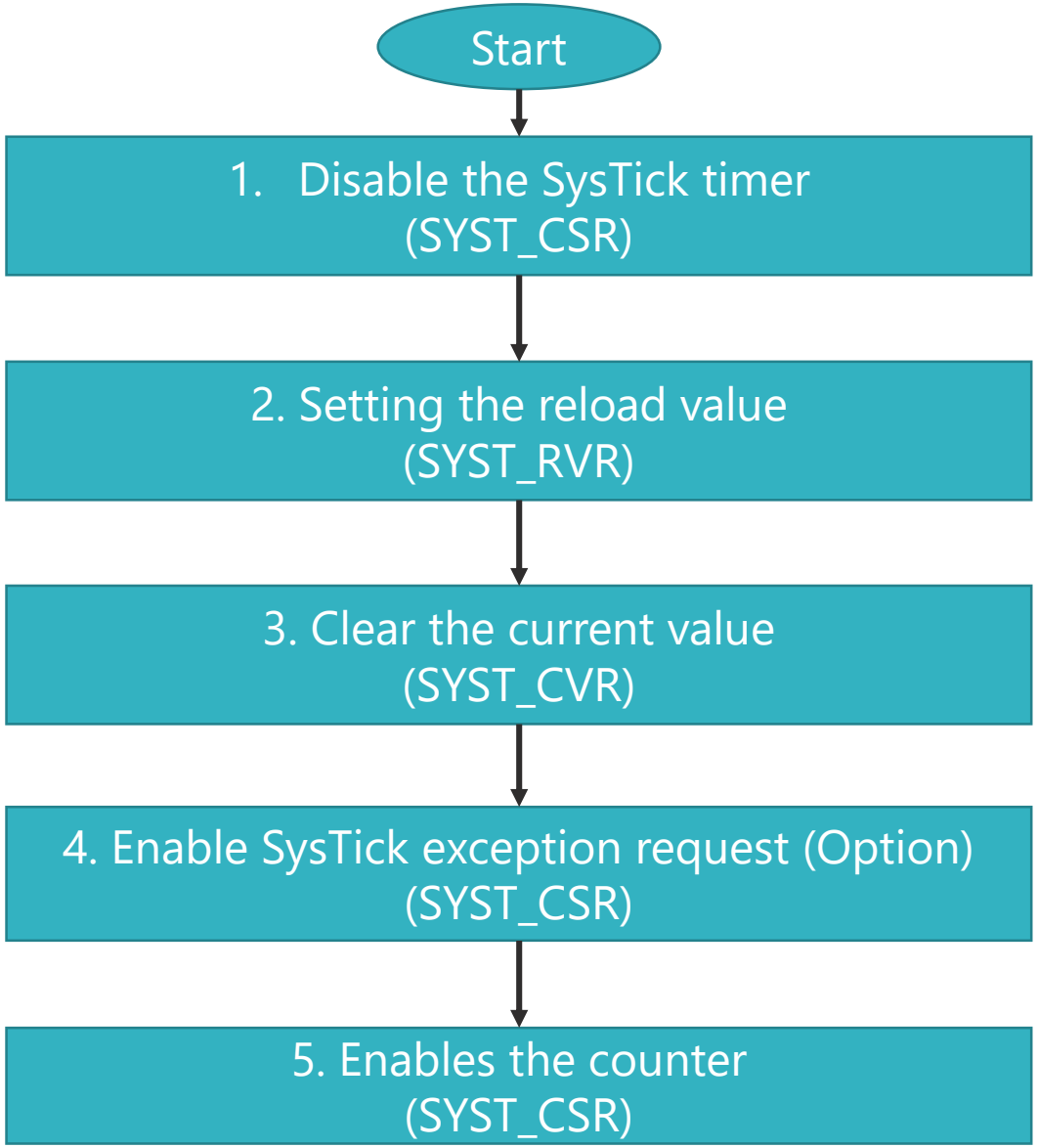
No.	Address	Name	Type	Description
1	0xE000E010	SYST_CSR	RW	Control and Status Register
2	0xE000E014	SYST_RVR	RW	Reload Value Register
3	0xE000E014	SYST_CVR	RW	Current Value Register
4	0xE000E01C	SYST_CALIB	RO	Calibration Value Register

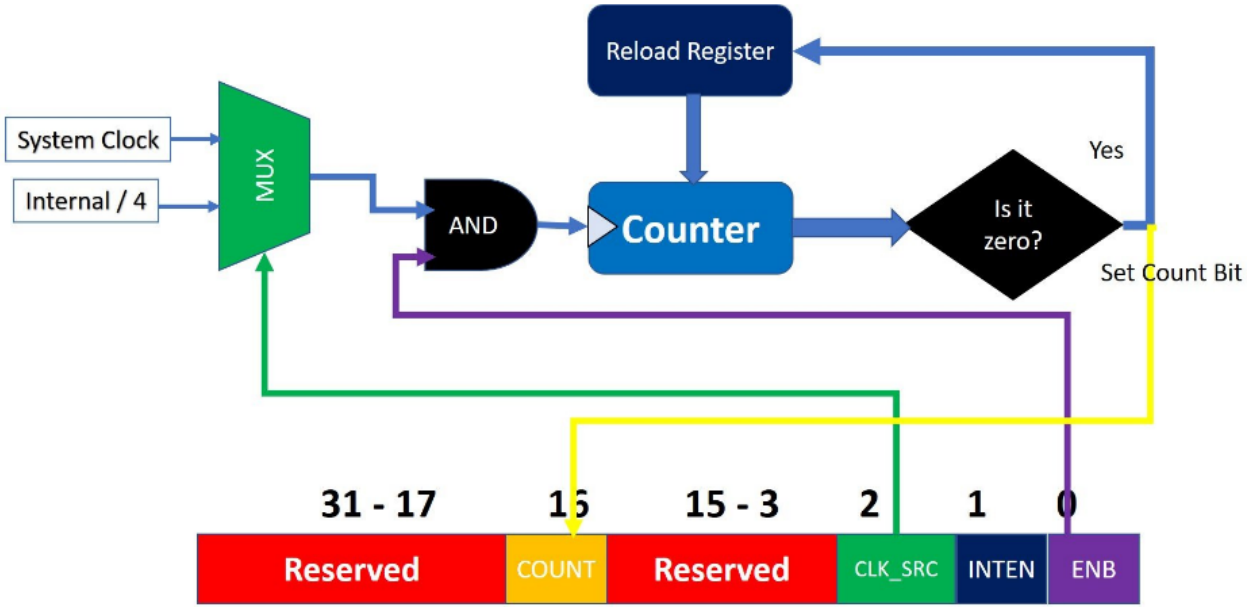
3. Operations of the SysTick timer

- The counter inside the SysTick is a **24-bit decrement counter**.
- It can decrement using the processor's clock, or using a reference clock signal
- When the counter is enabled by setting bit 0 of the Control and Status register, the current value register decrements at every processor clock cycle or every rising edge of the reference clock. If it reaches **zero**, it will then **load** the value from **the reload value register** and continue.



4. Configuration for SysTick timer





4.1. Configuration for periodic interrupt

Calculating the RELOAD value



- 24 bits, maximum value 0x00FF.FFFF(16,777,215)
- Counter counts down form RELOAD value to 0
- Writing RELOAD to 0: Disable Systick
- Time interval between 2 Systick interrupt:

$$\text{RELOAD} = \text{Interval(s)} * f_{CLKSOURCE}(\text{Hz}) - 1$$

4.2. Assignment 1

Blink Red Led every 500ms using polling technique

4.3. Assignment 2

Blink Red Led every 500ms using interrupt technique

4.4. Assignment 3

Write the delay function :

+ void delay(unsigned int ms);
+ ms: millisecond


Thank you